AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM275
Parallel and Distributed System

Final Examination
Summer Semester 2025/2026

Time Allowed: 2 Hours

AOU EGYPT

Instructions:

- Total Marks: **50**
- This exam consists of 3 parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are allowed.



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Part I: Multiple Choice Questions (10 Marks, 2 marks each)

1. What type of architecture does Cassandra use where every node can perform all read and write operations?
a) Master-slave architecture
b) Client-server architecture
c) Peer-to-peer distributed architecture
d) Centralized architecture

2. What is the main configuration file for Cassandra?
a) cassandra.conf
b) cassandra.yaml
c) cassandra.properties
d) cassandra.cfg

3. Which AWS service is equivalent to traditional on-premises routers and switches?
a) Amazon EC2
b) Amazon S3
c) Amazon VPC (Virtual Private Cloud)
d) Amazon RDS

4. In Cassandra data modeling, which approach is recommended?
a) Design tables first, then queries
b) Design queries first, then tables
c) Normalize first, then denormalize
d) Start with a relational model and convert

5. Which replication strategy is used when Cassandra is installed in only one data center?
a) NetworkTopologyStrategy
b) SimpleStrategy
c) MultiDCStrategy
d) RackAwareStrategy

Part II: Short Answer Questions (15 marks = 10+5)

1. Explain the CAP Theorem. What are its three components, and how does it apply to NoSQL databases like Cassandra? (10 marks).
2. Describe the AWS Global Infrastructure. Include the concepts of Regions, Availability Zones, and Data Centers. Why is it important to have multiple Availability Zones within a Region? (5 marks).

Part III: Answer The Following Questions (25 marks = 10+15)

A- Problem: Using CQL (Cassandra Query Language), perform the following tasks on a keyspaces named university:
1. create the keyspace with SimpleStrategy and replication factor 1. (3 marks)
2. Use the keyspace and create a table students with columns: student_id int PRIMARY KEY, name text, major text and gpa decimal (4 marks)
3. Insert the following three rows into the students table (101, 'Alice', 'CS', 3.8) (3 marks)

B- Answer the following questions:
1- What is the Gossip Protocol in Cassandra? How does it work and why is it important? (5 marks)
2- Explain the difference between Replication Factor and Replication Strategy in Cassandra. What are replication strategies used for? (5 marks)
3- What is the difference between a partition key and a clustering key in Cassandra? (5 marks)

End of questions

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